

Lumen — Portfolio Documentation

AI-Based Web Accessibility Checker

Repository: github.com/balisikh/ai-based-web-accessibility-checker · Local: <http://localhost:4376>

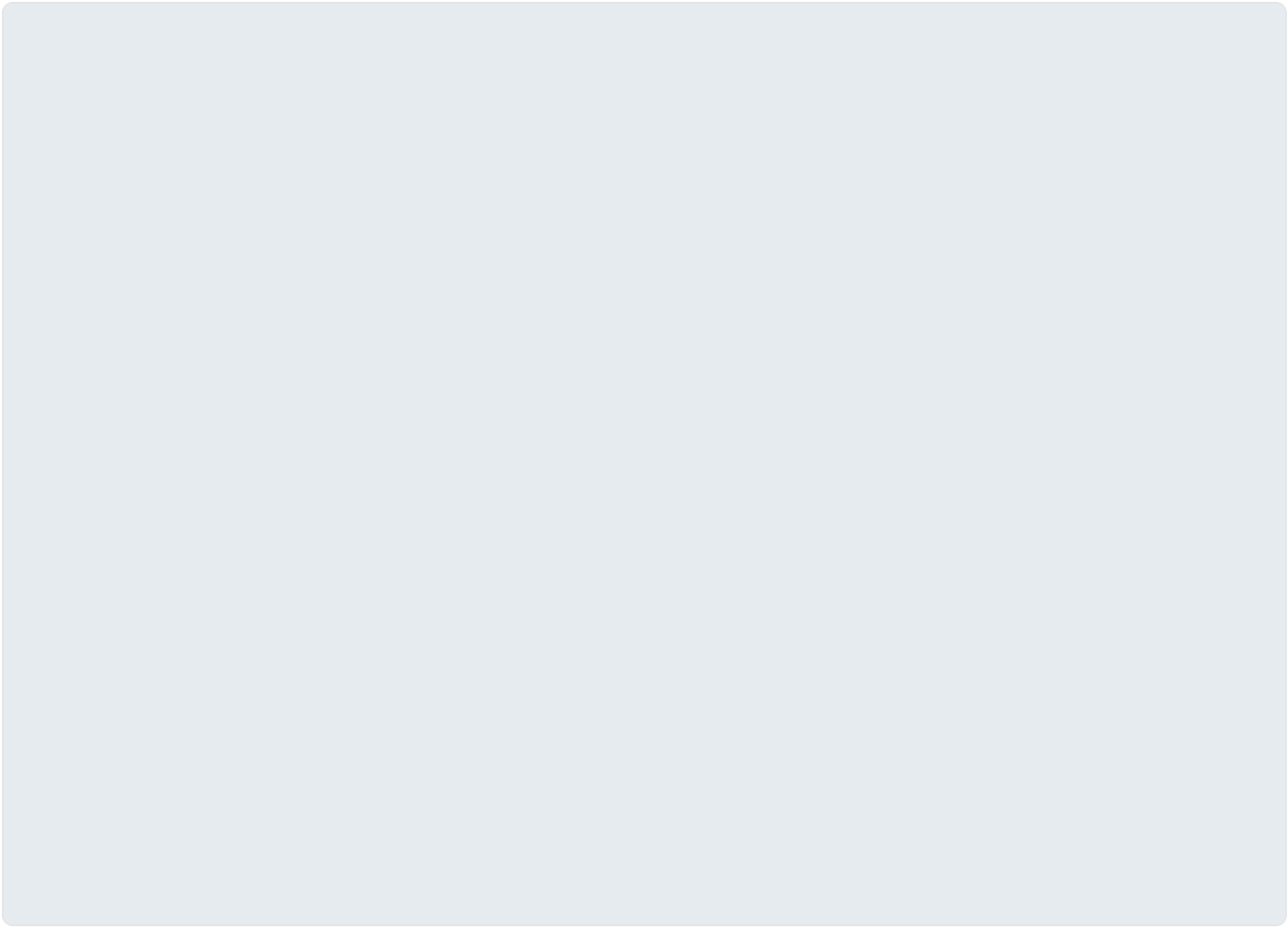
Assistive findings only — not a legal accessibility certificate or formal conformance audit.

Overview

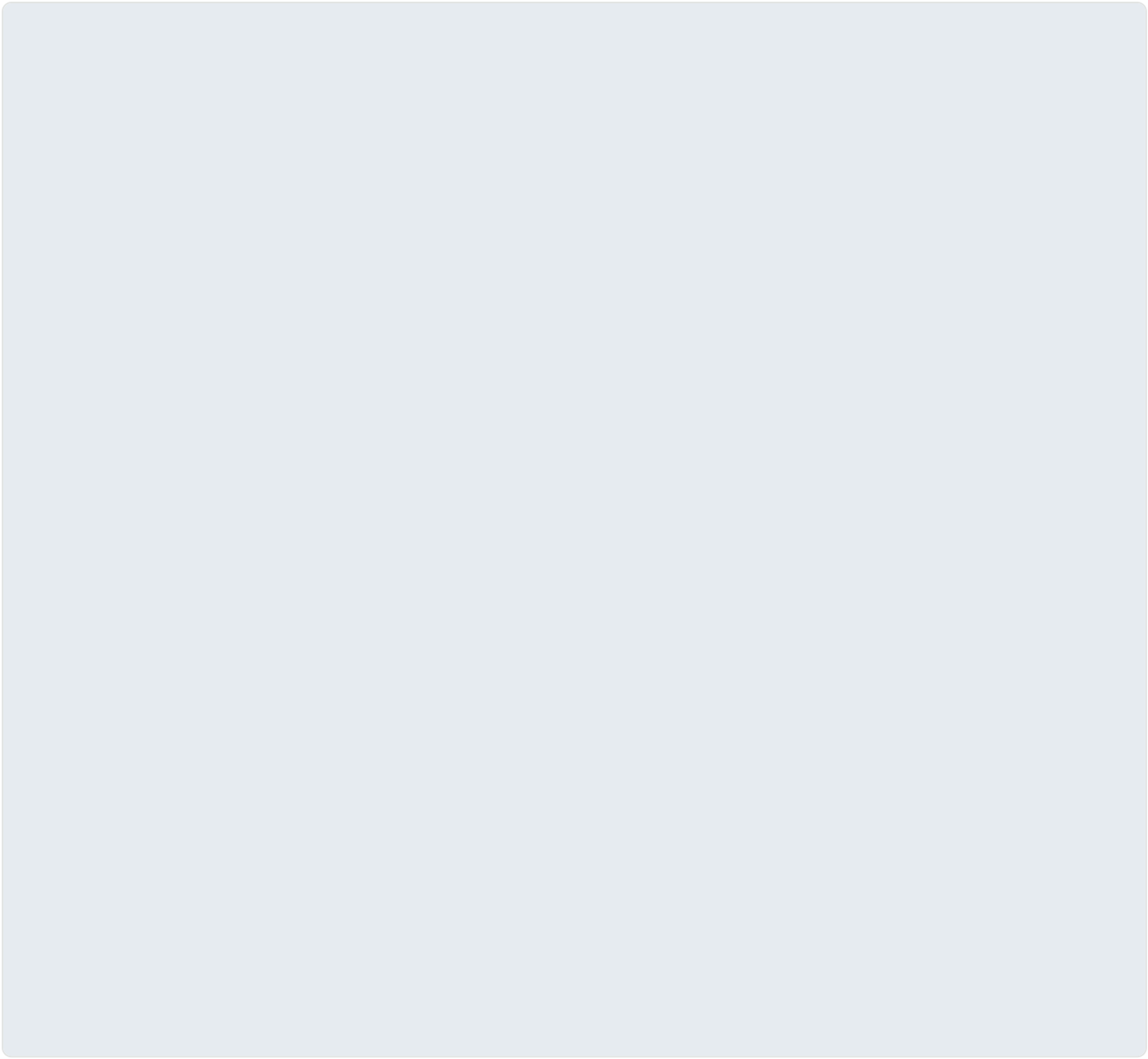
Lumen scans a public URL with Playwright + axe-core, returns a score, severities, rule help, optional AI tips, and JSON export.

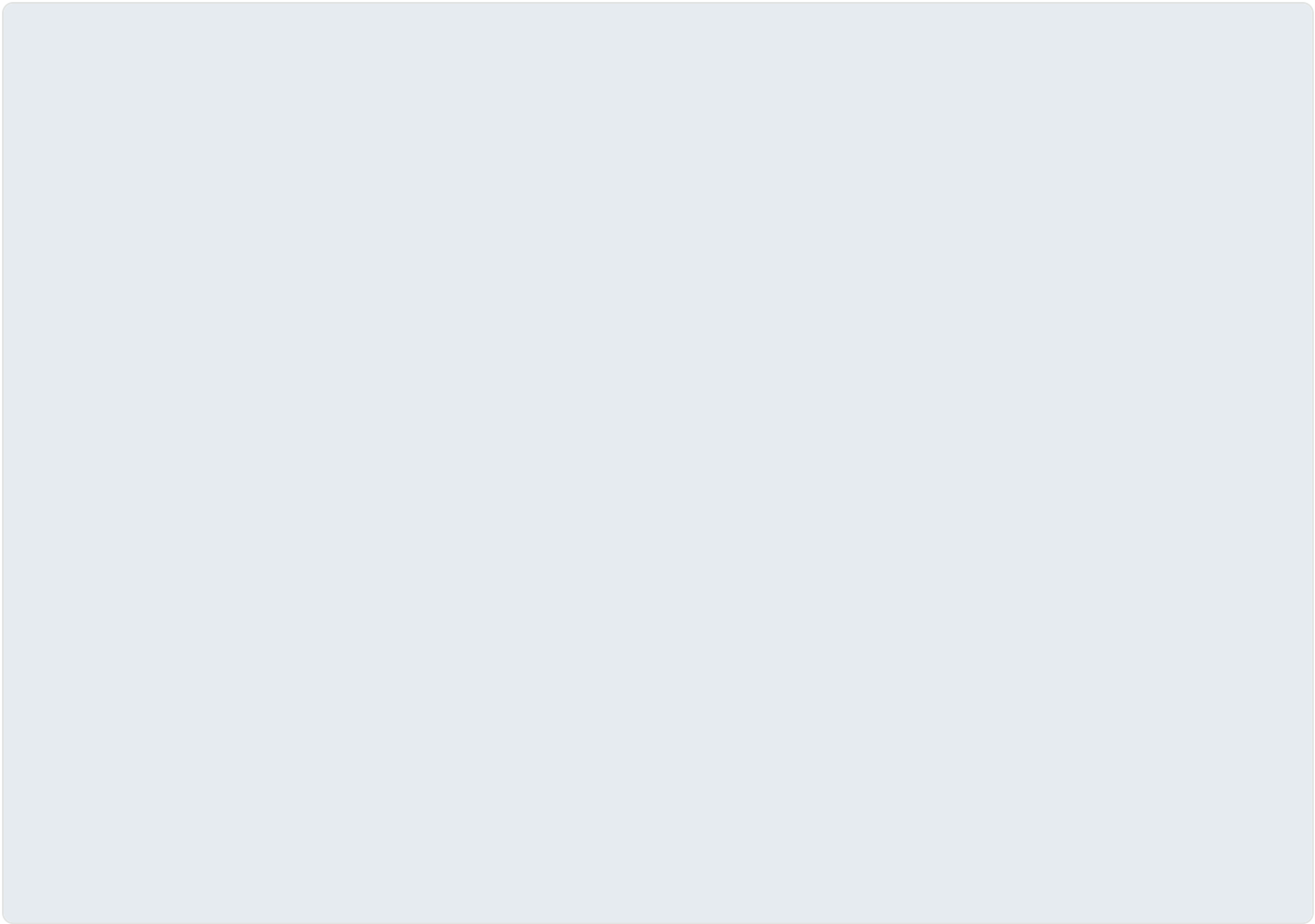
1. Paste a public website URL
2. Headless Chrome renders the page
3. axe-core checks WCAG A/AA-oriented rules
4. Score, severities, and issue details
5. Optional AI tips; JSON export

UI screenshots

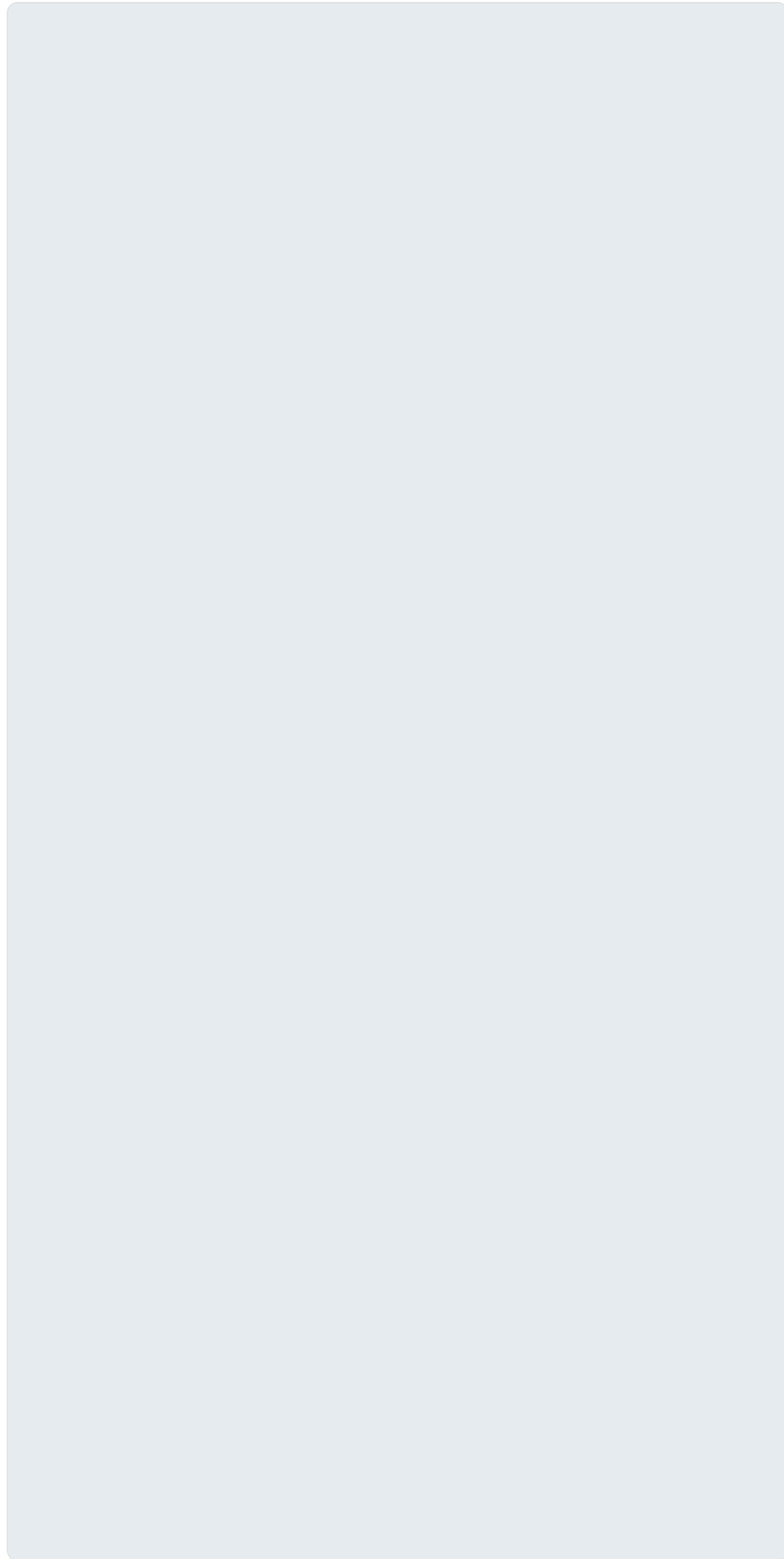


Batch results — 28 sites (overview)





Mobile home (390px)



Layout (light mode)

- **Header:** Lumen logo, Checker / Batch results, Light / Dark / System
- **Background:** Soft grey gradient + grid pattern
- **Card:** White panel with teal gradient strip on top
- **Home CTA:** Orange Check accessibility button

- **Batch CTA:** Teal Run a live scan button

Batch snapshot (2026-07-30)

Metric	Value
Websites tested	28
Passed	8 (4 clean · 4 with issues)
Failed	20
Total issues	374

Pass rule: Score ≥ 85 and Critical = 0.

Passed sites: Google UK, BBC iPlayer, BBC News, Disney+ UK, GitHub (balisikh), Google Maps, Wikipedia, example.com

MVP features

Area	Status
Scan UI, Playwright + axe	Done
Rate limiting, persistence, JSON export	Done
Optional AI tips, anonymous use	Done
PDF export / accounts / crawl	Not yet

Stack

- Next.js (React) + TypeScript
- Playwright (Chrome) + axe-core
- Postgres or PGlite
- Optional OpenAI-compatible AI tips

Local development note — plain-text UI incident (Aug 2026)

During local testing the app briefly showed **unstyled plain text** (e.g. nav labels run together). The design was not removed — **CSS failed to load**.

Symptom: White page, black text; HTML OK but stylesheet returned **500** from a stale/corrupt .next build on port 4376.

Cause: On low-memory Windows, Turbopack could crash compiling `globals.css`, leaving a broken build that still served HTML.

Fix (commit 541ec96):

- `dev / build` default to **webpack**
- `web/scripts/fix-styles.ps1` — stop server, delete `.next`, rebuild, restart
- Troubleshooting in `web/README.md` — confirm CSS status **200** in DevTools → Network

Resolution: After clean rebuild, UI matched screenshots again (header, gradient, card, batch table). Images in this PDF show the intended layout.

Generated 2026-08-03 from docs/screenshots and project README.